Yongheng Wang

Phone: (+86) 173-0661-8993 | Email: wangyh@eee.hku.hk | Website: hkuyonghengwang.github.io

EDUCATION

Tsinghua University Sep. 2021 – Jun. 2024
M.Phil. in Electrical Engineering GPA: 3.63/4.00

South China University of Technology Sep. 2017 – Jun. 2021
B.Eng. in Electrical Engineering and Automation GPA: 3.54/4.00

Imperial College London Aug. 2025 – Sep. 2025
Summer School

ACADEMIC APPOINTMENT

The University of Hong Kong, Research Assistant Sep. 2024 – Present

PUBLICATIONS

Journals

- **Y. Wang**, X. Shen, and Y. Xu. Joint Planning of Active Distribution Network and EV Charging Stations Considering Vehicle-to-Grid Functionality and Reactive Power Support. *CSEE Journal of Power and Energy Systems*, 2024. **Published.**
- **Y. Wang**, W. Gao, and X. Shen. Integrated Planning of Multi-Charging Infrastructure and Urban Distribution Networks Based on Smart Transportation Systems. *IEEE Transactions on Transportation Electrification*, 2026. **Under second review.**
- **Y. Wang**, X. Mo, and T. Liu. Tri-Level Two-Stage Stochastic-Robust Planning of Renewable Charging Stations and Distribution Networks: An Adaptive iC&CG Algorithm. *arXiv preprint* arXiv:2511.13776, 2025.
- **Y. Wang**, X. Mo, and T. Liu. Decentralized Dissipativity Certificates for Heterogeneous Power Systems. **Working paper.**
- W. Gao, **Y. Wang**, and X. Shen. Distributionally Robust Planning of PV-Storage-EV Stations and Low-Carbon Coastal City Distribution Network with i-C&CG Algorithm. *IEEE Transactions on Sustainable Energy*, 2025. **Under review.**
- H. Wang, X. Shen, and **Y. Wang**. Dynamic Reactive Power Optimization Based on Modified Generalized Benders Decomposition. *IEEE Transactions on Power Systems*, 2025. **Published.**
- C. Wei, **Y. Wang**, and X. Shen. Synergistic Planning of Photovoltaic Energy Storage-Charging Stations and Hydrogen Refueling Stations Considering Carbon Emission Flows. *Automation of Electric Power Systems*, 2023. **Published in Chinese.**

Conferences

- **Y. Wang**, R. Cui, and T. Liu. Two-Stage Robust Planning of Distribution Networks with Renewable Charging Stations: A Strong Optimization Framework. In *2026 IEEE PES International Meeting (PESIM)*, Hong Kong, China, 2026. **Best Paper.**
- **Y. Wang**, X. Mo, and T. Liu. Passivity-Based Decentralized Stability Criterion for Grid-Forming-Dominated Power Systems. In *Proceedings of the 2026 IEEE 65th Conference on Decision and Control (CDC)*. **Under review.**
- G. Liu, **Y. Wang**, et al. Coordinated Planning of Active Distribution Network and V2G Charging Stations Considering the Load Characteristics of V2G Stations. In *2022 IEEE 6th Conference on Energy Internet and Energy System Integration (EI^2)*, Chengdu, China, 2022. **Published.**
- G. Liu, W. Chen, **Y. Wang**, et al. Co-Planning of ADN and EV Charging Stations Considering EV Spatial Migration and Sequential Charging Characteristics. In *2023 8th Asia Conference on Power and Electrical Engineering (ACPEE)*, Tianjin, China, 2023. **Published.**

Patents

- X. Shen, **Y. Wang**, et al. Method for Joint Planning of Active Distribution Network and V2G Charging Stations. Chinese Patent 202310630383.X, 2023.
- G. Liu, W. Zheng, **Y. Wang**, et al. Experimental Device for Simulating Different Contact States of Plum Blossom Contacts by Adjusting the Insertion Depth of Static Contacts. Chinese Patent ZL201911315956.X, 2021.
- X. Shen, W. Chen, **Y. Wang**, et al. Method for Collaborative Planning of New Energy Vehicle Charging Stations Considering Carbon Emission Flow. Chinese Patent 202311022600.3, 2023.
- W. Tang, Y. Zhao, C. Zhong, X. Zhao, X. Shen, **Y. Wang**, et al. Method for Optimal Location and Sizing of Wind, Solar, and V2G Charging Stations in Distribution Networks Based on Improved Beetle Antennae Search Particle Swarm Algorithm. Chinese Patent 35082119900201004.X, 2022.

RESEARCH PROJECTS

National Natural Science Foundation of China Sep. 2024 – Present
Dissipativity-Based Distributed Event-Triggered Control of Complex Dynamic Networks and Its Applications to Microgrid Control *Key Researcher*

Research Grants Council of the Hong Kong Special Administrative Region (Early Career Scheme) Sep. 2024 – Present
Small-Signal Stability Analysis of Power Systems with High Penetration of Converter-Interfaced Generation *Key Researcher*

China Southern Power Grid Corp. Jan. 2022 – Nov. 2024
Key Technology and Demonstration for Large-Scale Electric Vehicle Interactions with the Power Grid *Key Researcher*

National Natural Science Foundation of China Sep. 2021 – Dec. 2023
Research on Data-Driven Planning Method for Integrated Energy Distribution System Considering Multiple Energy Storage *Key Researcher*

National Natural Science Foundation of China Sep. 2020 – Jun. 2021
Damage Mechanism of Lightning Current on Tensioned Stranded Metallic Ground Wires *Participant*

PROFESSIONAL EXPERIENCE

China Southern Power Grid Company Limited Apr. 2022 – Dec. 2022
Intern, Shenzhen Power Supply Bureau

China Southern Power Grid Company Limited Jun. 2018 – Sep. 2018
Intern, Guangzhou Power Supply Bureau

HONORS AND AWARDS

- **National Scholarship** (awarded twice, Top 2%) 2018–2019; 2019–2020
- **Outstanding Graduate**, Tsinghua University (Top 2%) 2023–2024
- **Outstanding Graduation Thesis**, Tsinghua University (Top 5%) 2023–2024
- **Best Poster Award**, International Workshop on Learning and Information Theory (Top 2%) 2023
- **First Prize Scholarship**, Tsinghua University (Top 5%) 2022–2023
- **Outstanding Graduate**, South China University of Technology (Top 5%) 2021–2022
- **Third Prize in Professional Practice**, Tsinghua University (Top 10%) 2024
- **Kang Dewei Innovation Scholarship**, South China University of Technology (Top 10%) 2017–2018
- **Service Honors**, South China University of Technology: Outstanding Student Leader (2019–2020), Outstanding Intern in Power Grid (2018–2019), Outstanding Member of Student Union (2018–2019)

LEADERSHIP AND SERVICE

Tsinghua University Student Union Mar. 2022 – Dec. 2022
Member, Practice Department

Tsinghua International Graduate School Oct. 2021 – Aug. 2022
Monitor, Electrical Engineering Class 21

South China University of Technology Student Union May 2018 – Jul. 2019
Secretary, Department of Manpower and Liaison

Student Innovation Club of South China University of Technology Mar. 2018 – Aug. 2018
Member, Outreach Practice Department

Art Group of South China University of Technology Jul. 2017 – Jun. 2019
Leader, Host Team

SKILLS

Programming	MATLAB, Python, C++
Software	Microsoft Office, LaTeX, Photoshop
Languages	English, Chinese

Updated April 2026